

5(a)(i)	$\left[R = \frac{\rho L}{A} \right]$ Cross-sectional area = $\pi \left(\frac{0.18 \times 10^{-3}}{2} \right)^2 = 2.545 \times 10^{-8} \text{ m}^2$ Resistance = $\frac{(1.7 \times 10^{-8})(96)}{2.545 \times 10^{-8}} = 64.13 = 64 \text{ } \Omega$
5(a)(ii)	Assuming the volume of the wire remains constant, its length increases and its cross-sectional area decreases. Since its resistance is proportional to its length and inversely proportional to its cross-sectional area, its resistance increases.
5(a)(iii)	With 16 strands in parallel, the cross-sectional area is increased by a factor of 16. Therefore the resistance will be decreased by a factor of 16. Resistance = $\frac{64.13}{16} = 4.008 = 4.0 \text{ } \Omega$
5(b)(i)	$\left[P = I^2 R \right]$ Power = $(2.5)^2 (4.008) = 25.05 = 25 \text{ W}$
5(b)(ii)	$\left[I = nAqv \right]$ $\frac{2.5}{16} = (8.5 \times 10^{28})(\pi)(0.090 \times 10^{-3})^2 (1.6 \times 10^{-19})v$ $v = 4.51 \times 10^{-4} \text{ m s}^{-1}$
6(a)	Due to wave particle duality, the electrons possess wave properties with de Broglie